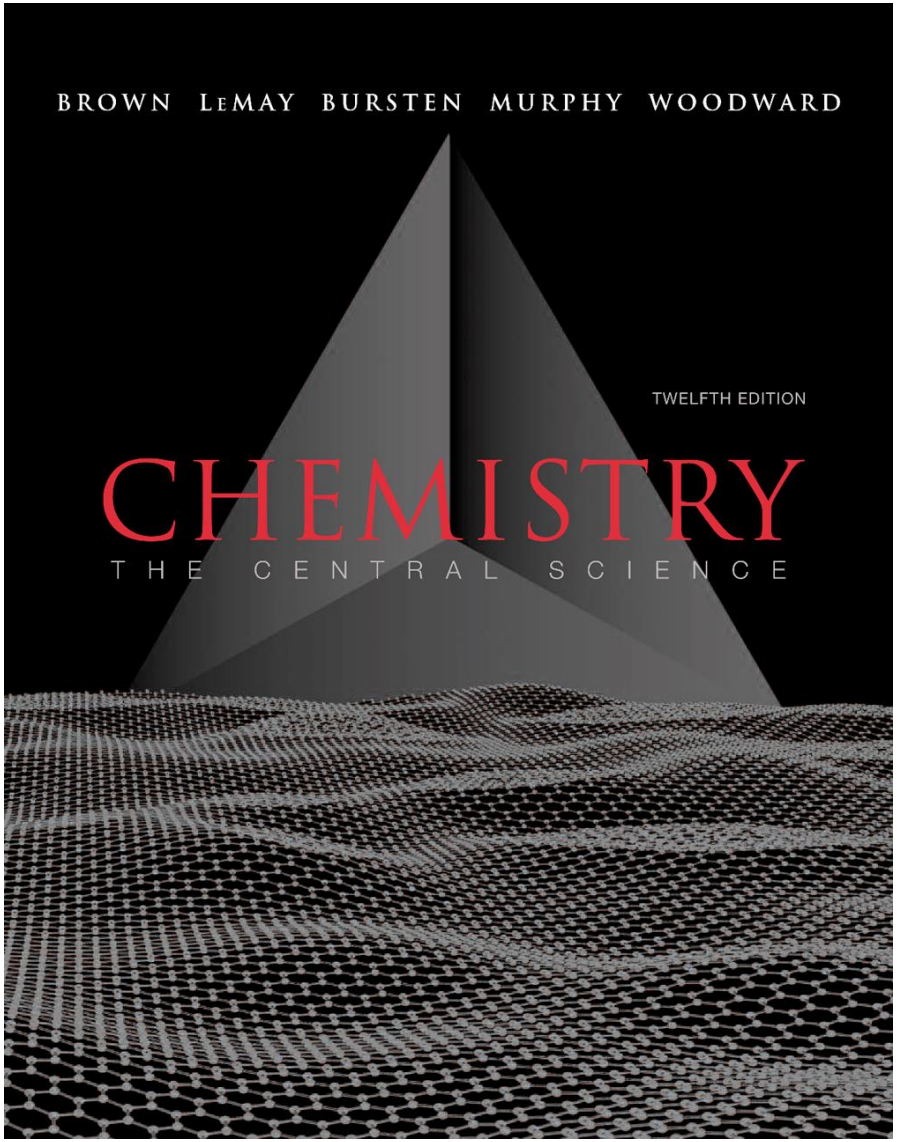


Lecture Presentation

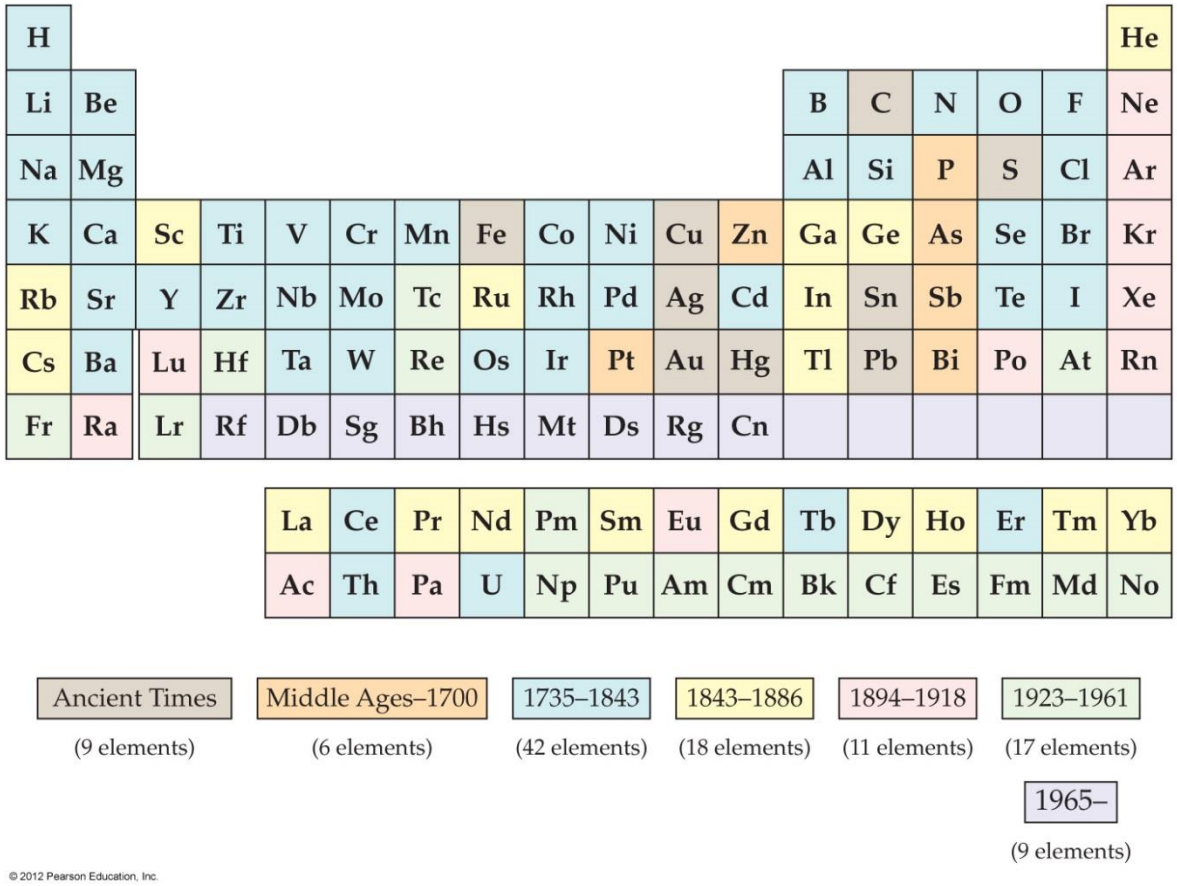
Chapter 7

Periodic Properties of the Elements

John D. Bookstaver
St. Charles Community College
Cottleville, MO

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Development of Periodic Table



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Dmitri Mendeleev and Lothar Meyer independently came to the same conclusion about how elements should be grouped.

Periodic Properties of the Elements

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Development of Periodic Table

TABLE 7.1 • Comparison of the Properties of Eka-Silicon Predicted by Mendeleev with the Observed Properties of Germanium

Property	Mendeleev's Predictions for Eka-Silicon (made in 1871)	Observed Properties of Germanium (discovered in 1886)
Atomic weight	72	72.59
Density (g/cm^3)	5.5	5.35
Specific heat (J/g-K)	0.305	0.309
Melting point (°C)	High	947
Color	Dark gray	Grayish white
Formula of oxide	XO_2	GeO_2
Density of oxide (g/cm^3)	4.7	4.70
Formula of chloride	XCl_4	$GeCl_4$
Boiling point of chloride (°C)	A little under 100	84

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Mendeleev, for instance, predicted the discovery of germanium (which he called eka-silicon) as an element with an atomic weight between that of zinc and arsenic, but with chemical properties similar to those of silicon.

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In 1913, two years after Rutherford proposed the nuclear model of the atom, English physicist Henry Moseley (1887–1915) developed the concept of atomic numbers. Bombarding different elements with high-energy electrons, Moseley found that each element produced X rays of a unique frequency and that the
frequency generally increased as the atomic mass increased. He arranged the X-ray frequencies in order by assigning a unique whole number, called an atomic number, to
each element. Moseley correctly identified the atomic number as the number of protons in the nucleus of the atom.

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- ✓ The periodic nature of the periodic table arises from repeating patterns in the electron configurations of the elements. Elements in the same column contain the same number of electrons in their **valence orbitals—**the occupied orbitals that hold the electrons involved in bonding. For example, **O** [He] $2s^2 2p^4$ and **S** [Ne] $3s^2 3p^4$ are both in group 6A. The similarity of the electron distribution in their valence s and p orbitals leads to similarities in the properties of these two elements.

- ✓ Electron configurations can be used to explain differences as well as similarities in the properties of elements. Despite similarities in their electron distributions, elemental oxygen and sulfur differ in fundamental ways. For example, at room temperature oxygen is a colorless gas but sulfur is a yellow solid.

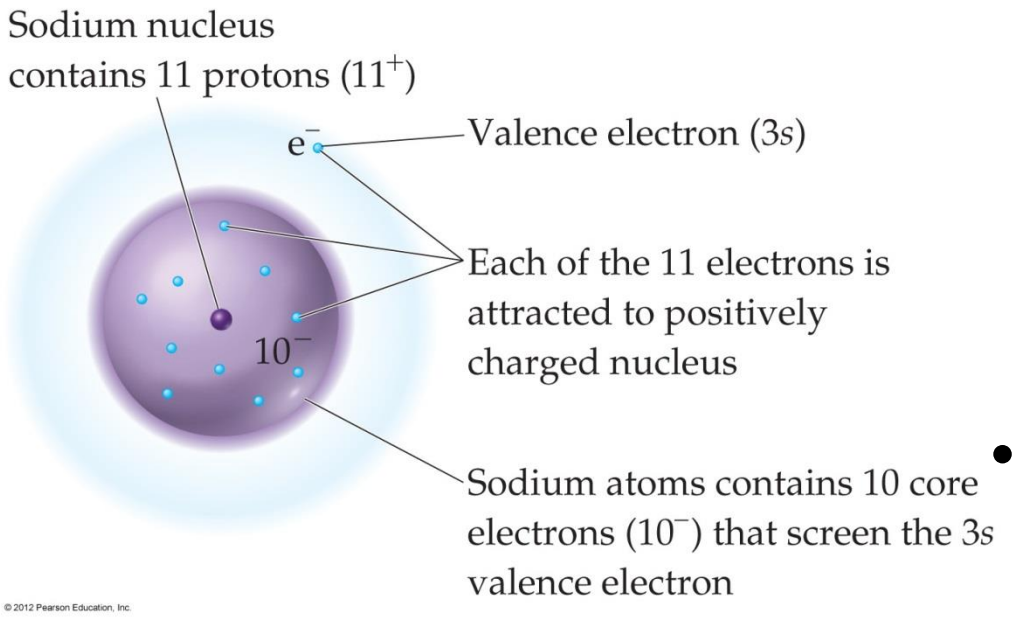
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Periodic Trends

- In this chapter, we will rationalize observed trends in
  - Sizes of atoms and ions.
  - Ionization energy.
  - Electron affinity.

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Effective Nuclear Charge

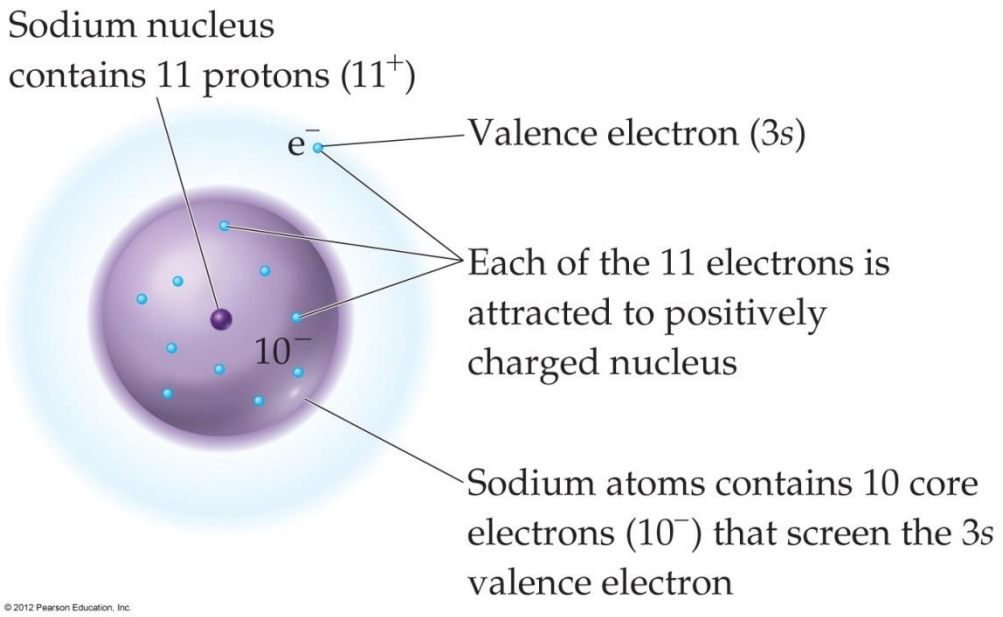


- In a many-electron atom, electrons are both attracted to the nucleus and repelled by other electrons.
- The nuclear charge that an electron experiences depends on both factors.

Periodic Properties of the Elements

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Effective Nuclear Charge



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The effective nuclear charge, Z_{eff}, is found this way:

$$Z_{eff} = Z - S$$

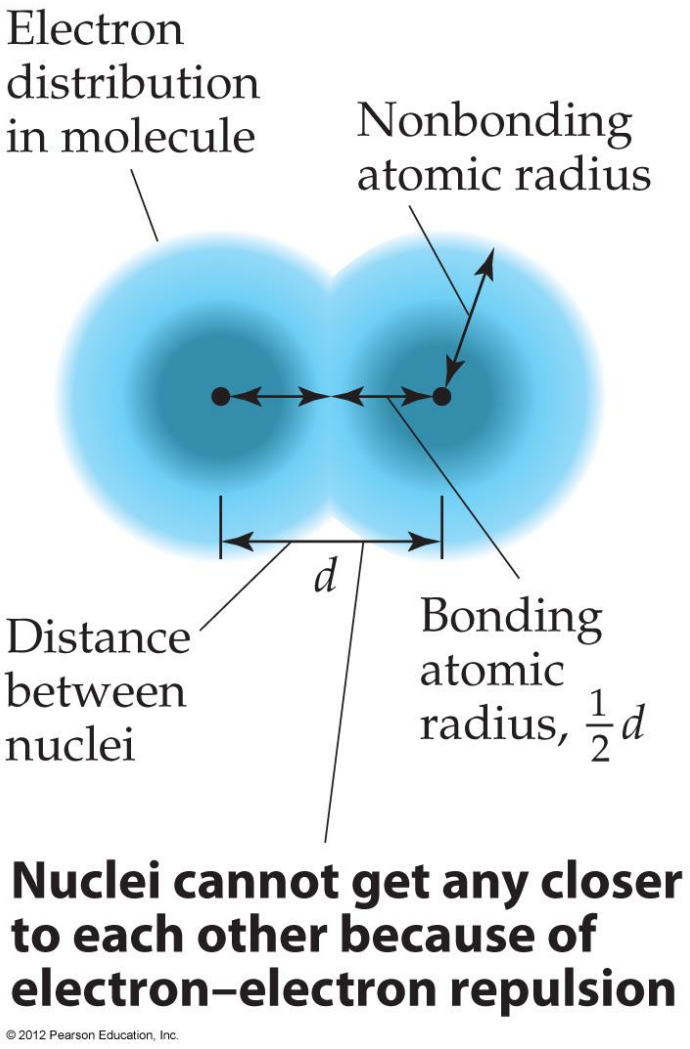
where Z is the atomic number and S is a screening constant, usually close to the number of inner electrons.

Periodic Properties of the Elements

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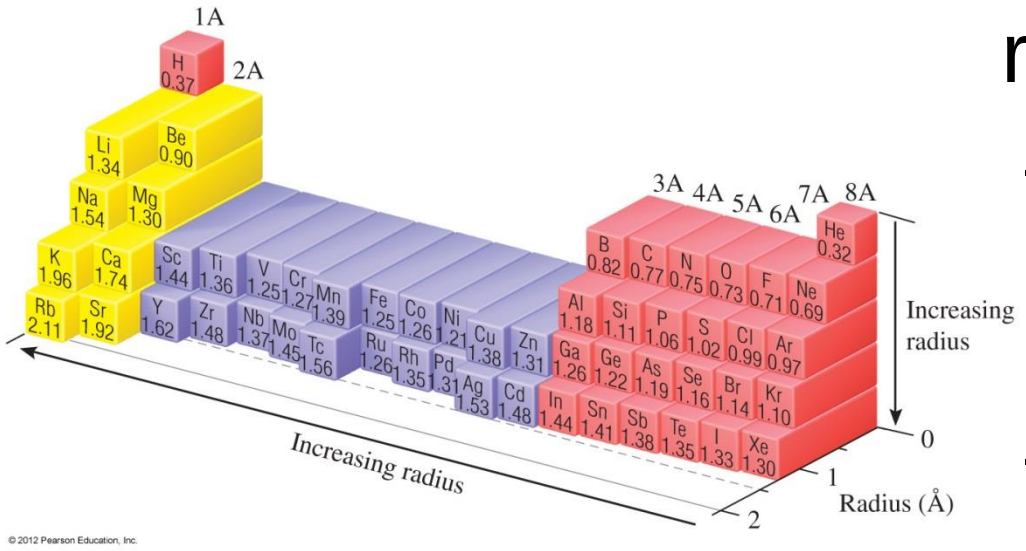
What Is the Size of an Atom?

The **bonding atomic radius** is defined as one-half of the distance between covalently bonded nuclei.



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Sizes of Atoms



The bonding atomic radius tends to

— Decrease from left to right across a row

(due to increasing Z_{eff}).

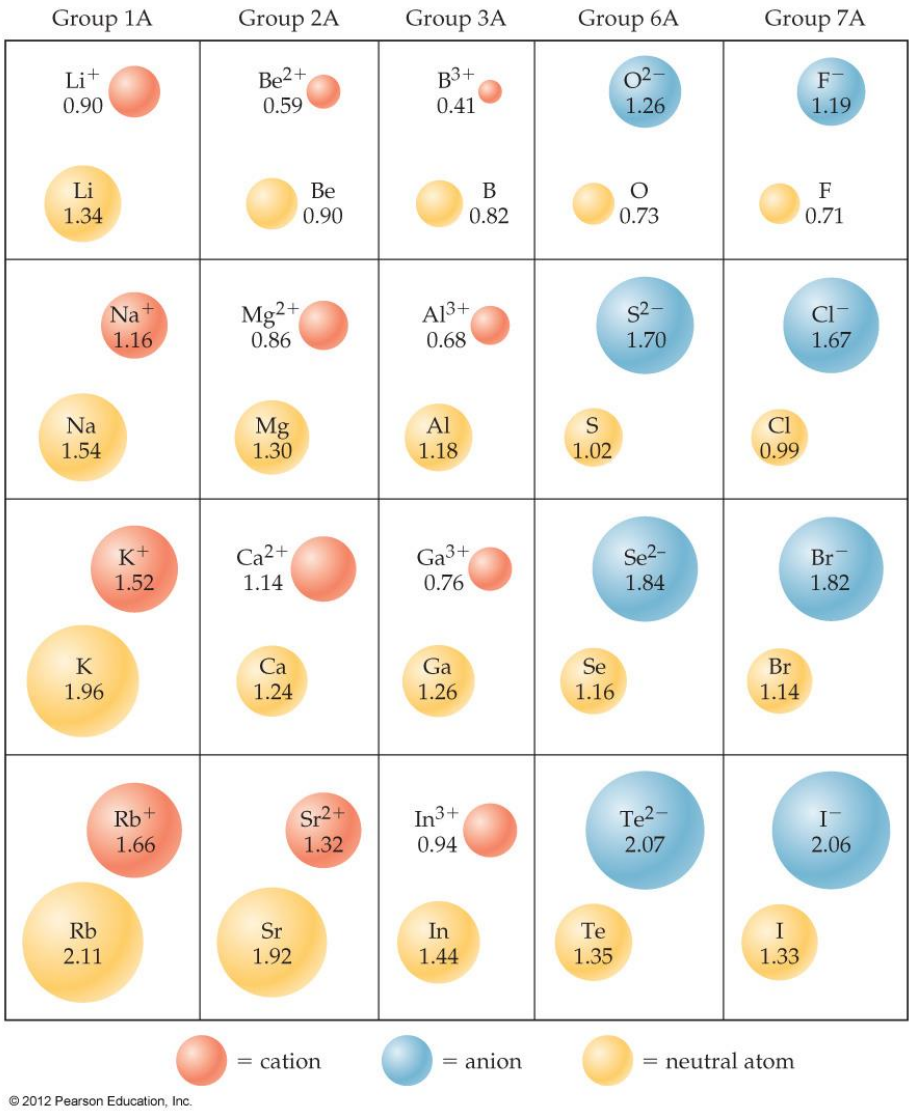
— Increase from top to bottom of a column

(due to the increasing value of *n*).

Periodic Properties of the Elements

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Sizes of Ions

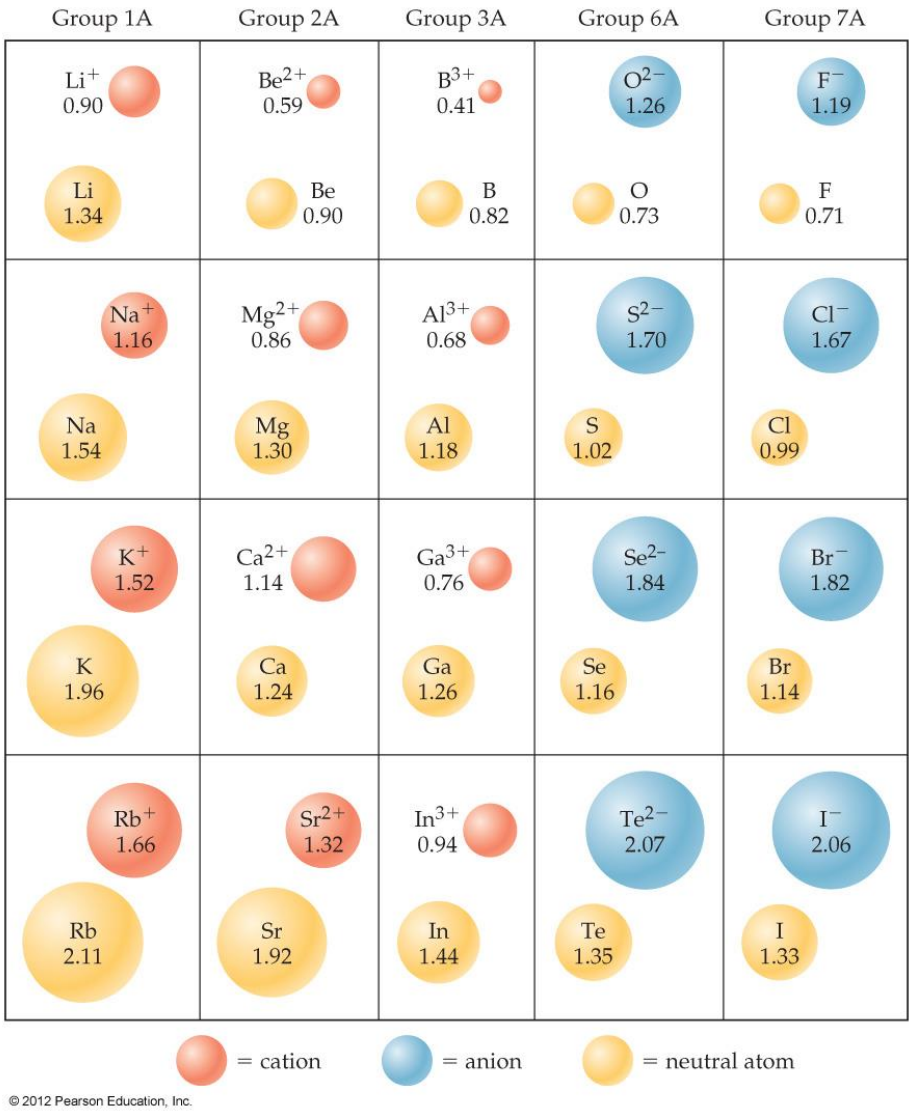


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- Ionic size depends upon
  - The nuclear charge.
  - The number of electrons.
  - The orbitals in which electrons reside.

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Sizes of Ions



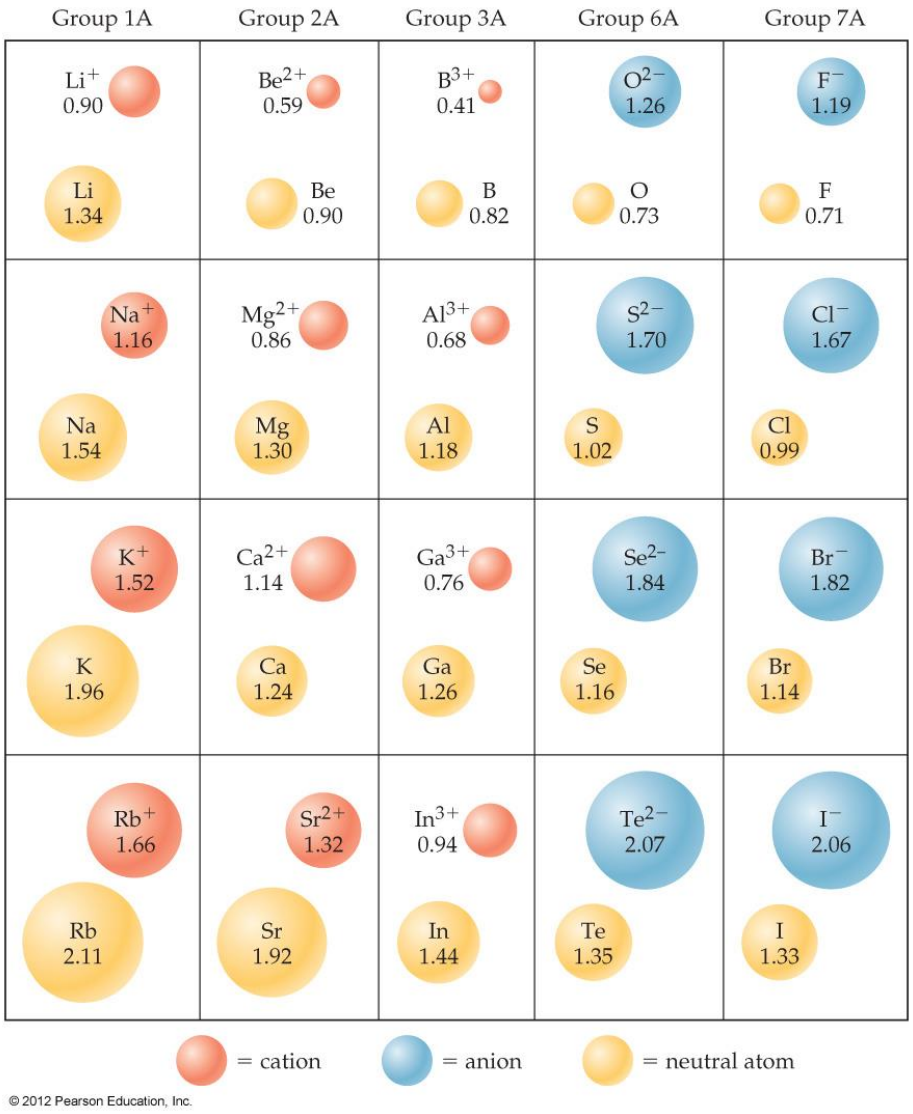
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- Cations are smaller than their parent atoms:
  - The outermost electron is removed and repulsions between electrons are reduced.

Periodic Properties of the Elements

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Sizes of Ions



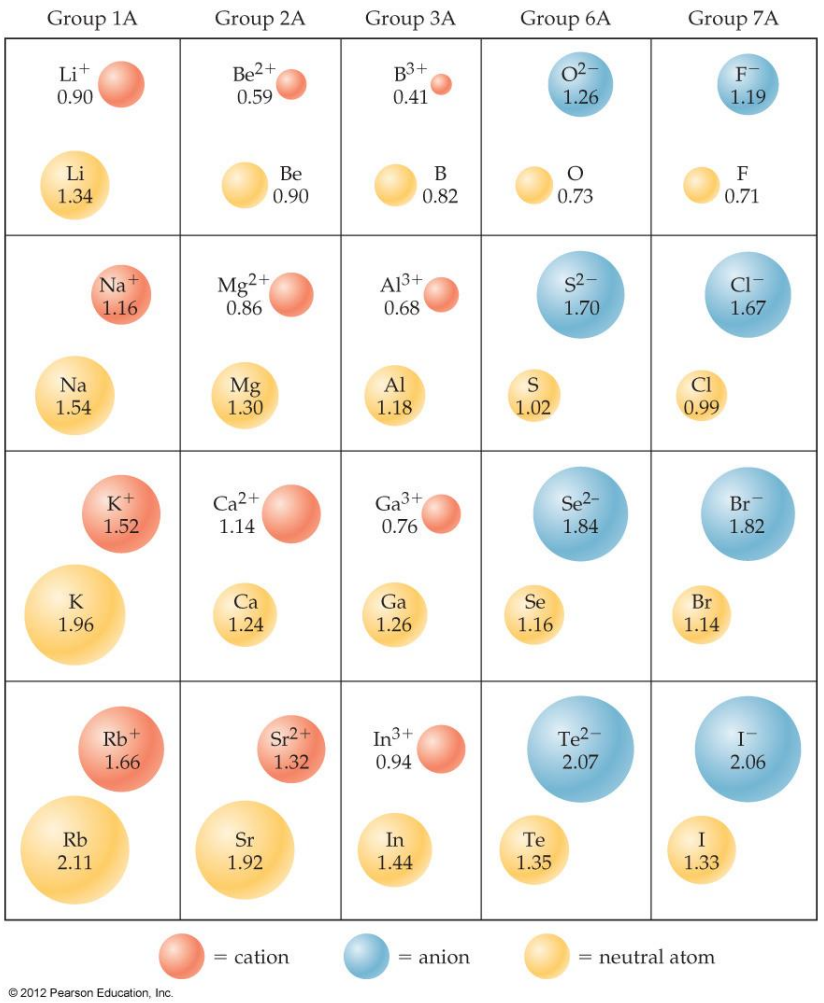
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- Anions are larger than their parent atoms"
  - Electrons are added and repulsions between electrons are increased.

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Sizes of Ions

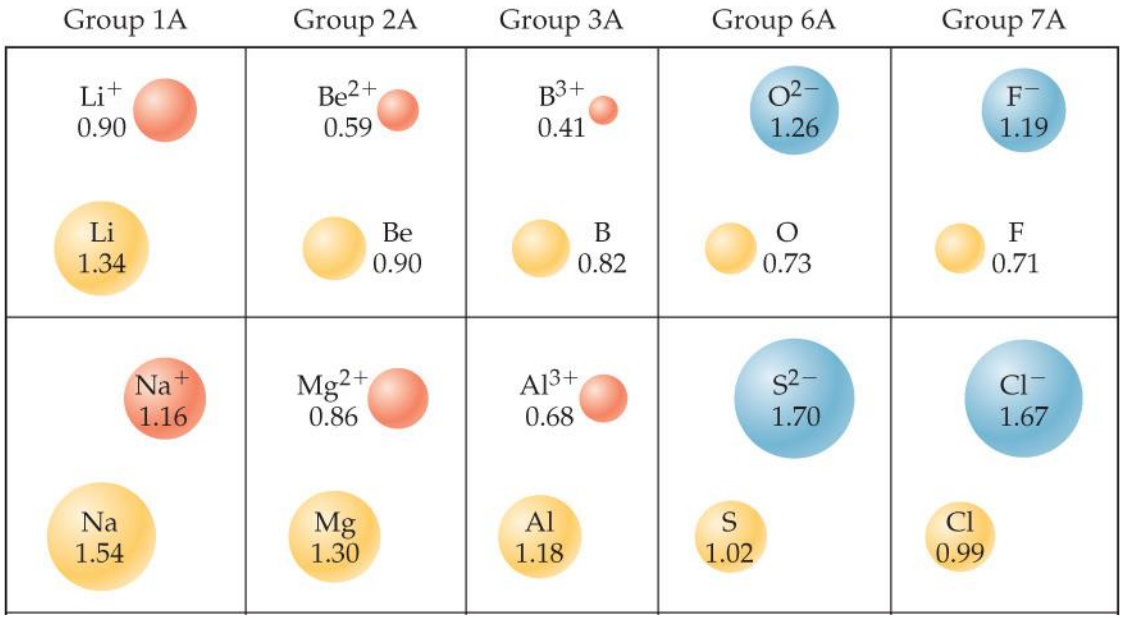
- Ions increase in size as you go down a column:
  - This increase in size is due to the increasing value of *n*.



Periodic Properties of the Elements

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Sizes of Ions



- In an **isoelectronic series**, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.

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An **isoelectronic series** is a group of ions all containing the same number of electrons. For example, each ion in the isoelectronic series O^{2-}, F^-, Na^+, Mg^{2+}, and Al^{3+} has 10 electrons. In any isoelectronic series we can list the members in order of increasing atomic number; therefore, the nuclear charge increases as we move through the series. Because the number of electrons remains constant, ionic radius decreases with increasing nuclear charge as the electrons are more strongly attracted to the nucleus:

—Increasing nuclear charge ⟶

O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
1.26 Å	1.19 Å	1.16 Å	0.86 Å	0.68 Å

—Decreasing ionic radius ⟶

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Ionization Energy

- The **ionization energy** is the amount of energy required to remove an electron from the ground state of a gaseous atom or ion.
  - The first ionization energy is that energy required to remove the first electron.
  - The second ionization energy is that energy required to remove the second electron, etc.

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Ionization Energy

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, the ionization energy takes a quantum leap.

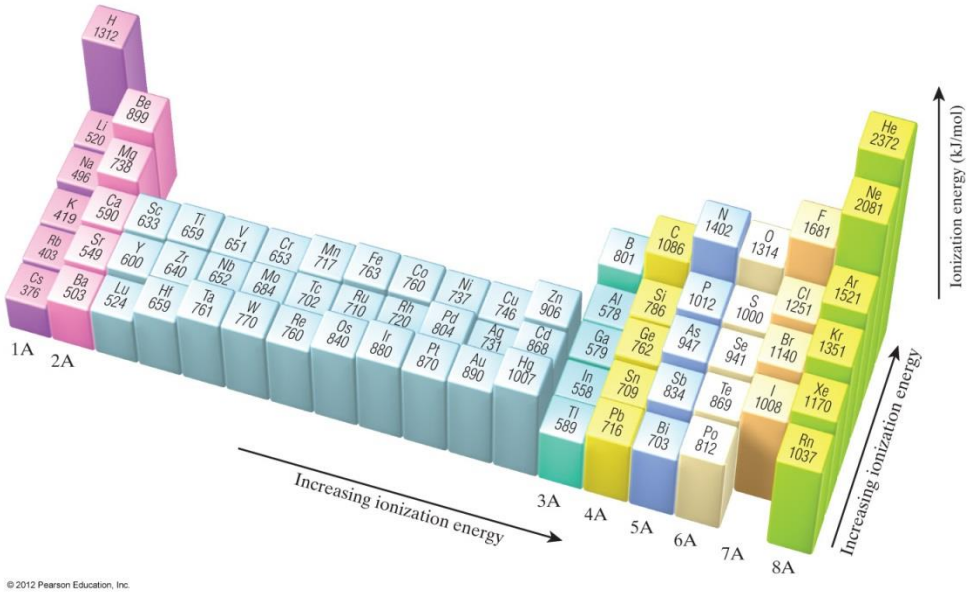
TABLE 7.2 • Successive Values of Ionization Energies, *I*, for the Elements Sodium through Argon (kJ/mol)

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	496	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

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Trends in First Ionization Energies



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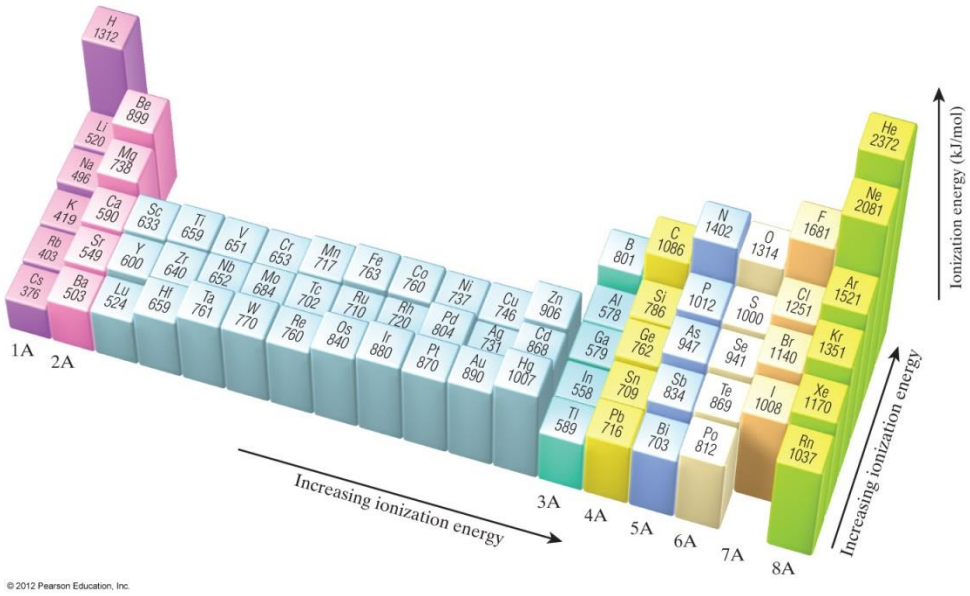
- As one goes down a column, less energy is required to remove the first electron.
  - For atoms in the same group, Z_{eff} is essentially the same, but the valence electrons are farther from the nucleus.

Periodic Properties of the Elements

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Trends in First Ionization Energies

- Generally, as one goes across a row, it gets harder to remove an electron.
  - As you go from left to right, Z_{eff} increases.

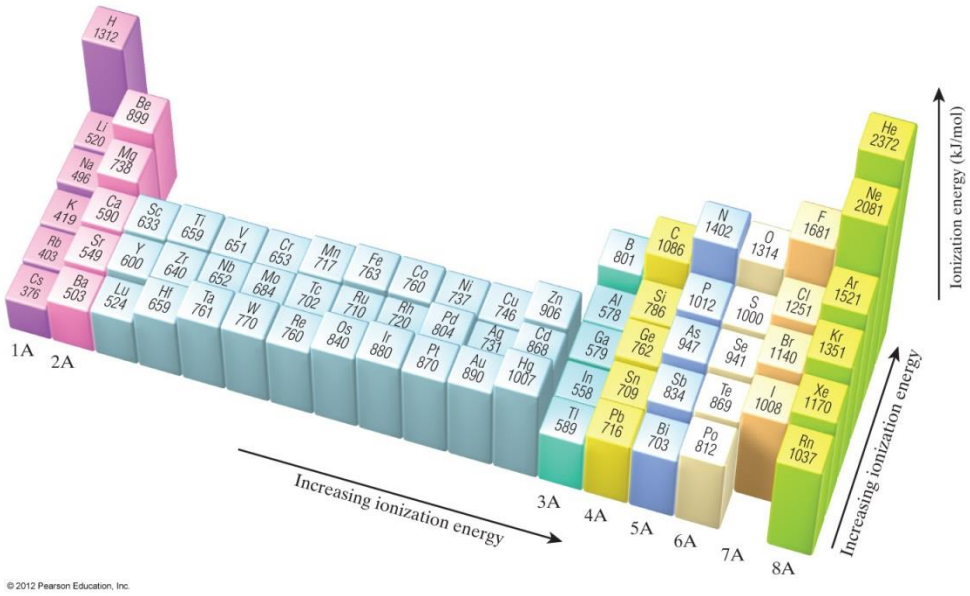


Periodic Properties of the Elements

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Trends in First Ionization Energies

However, there are two apparent discontinuities in this trend.

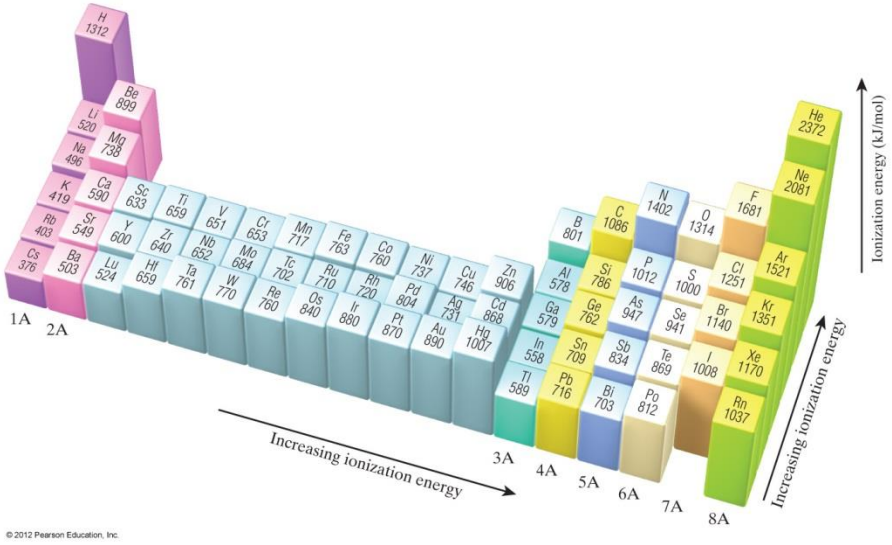


Periodic Properties of the Elements

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Trends in First Ionization Energies

- The first occurs between Groups IIA and IIIA.
- In this case the electron is removed from a *p* orbital rather than an *s* orbital.
  - The electron removed is farther from the nucleus.
  - There is also a small amount of repulsion by the *s* electrons.

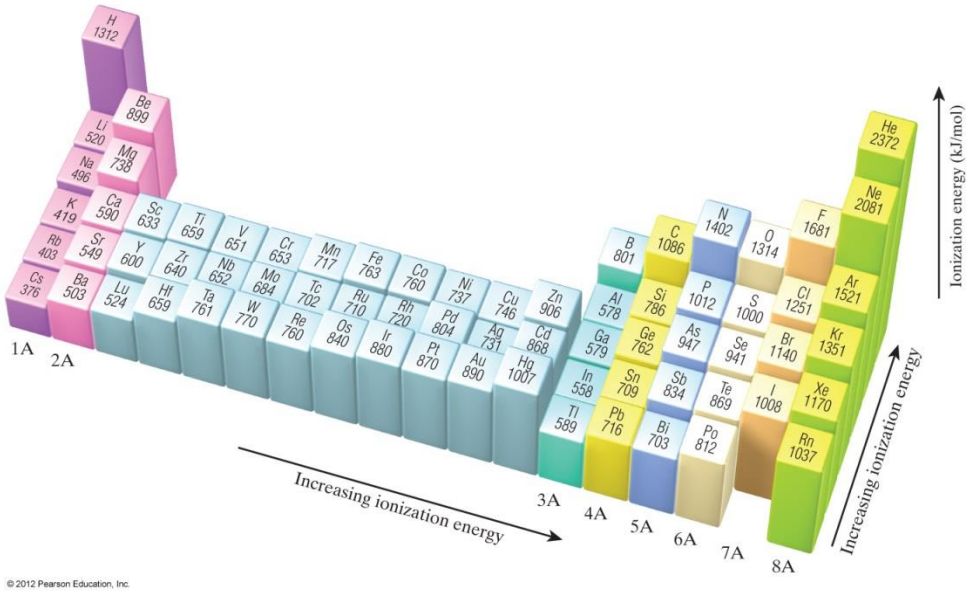


Periodic Properties of the Elements

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Trends in First Ionization Energies

- The second discontinuity occurs between Groups VA and VIA.
  - The electron removed comes from a doubly occupied orbital.
  - Repulsion from the other electron in the orbital aids in its removal.



Periodic Properties of the Elements

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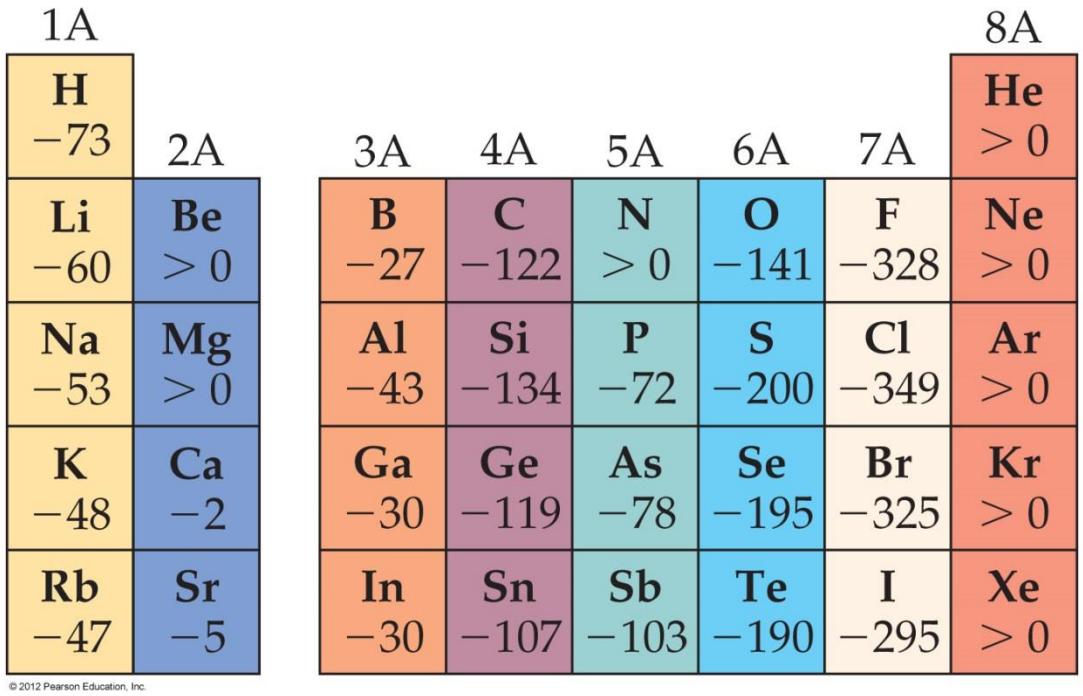
Electron Affinity

Electron affinity is the energy change accompanying the addition of an electron to a gaseous atom:

$$Cl + e^- \longrightarrow Cl^-$$

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Trends in Electron Affinity



1A	2A	3A	4A	5A	6A	7A	8A
H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0

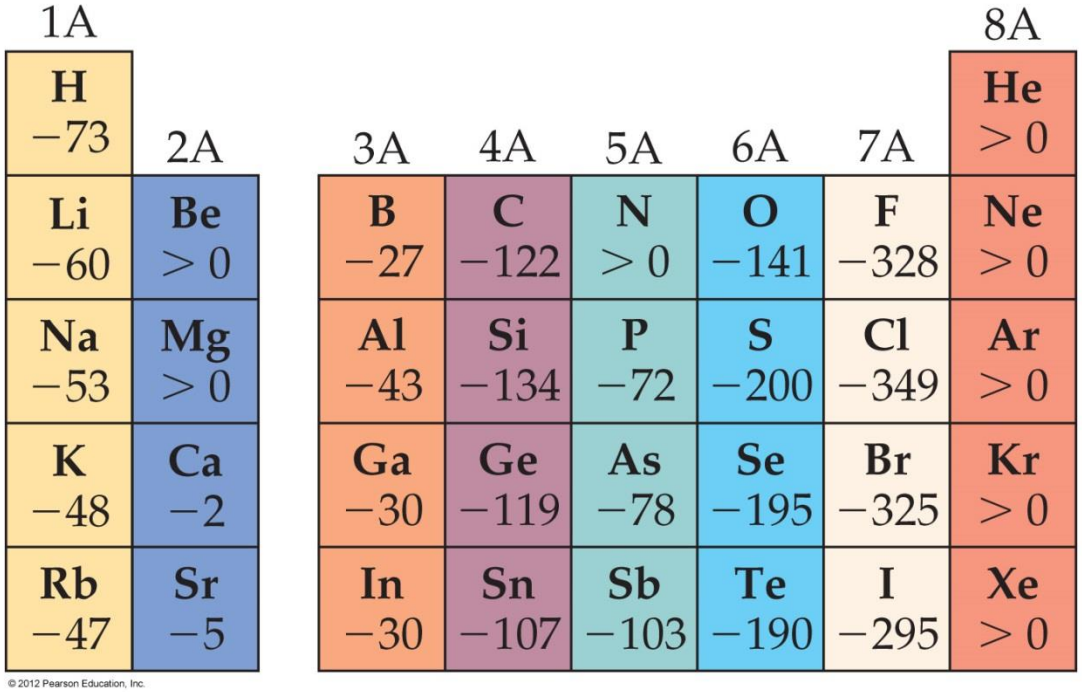
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In general, electron affinity becomes more exothermic as you go from left to right across a row.

Periodic Properties of the Elements

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Trends in Electron Affinity



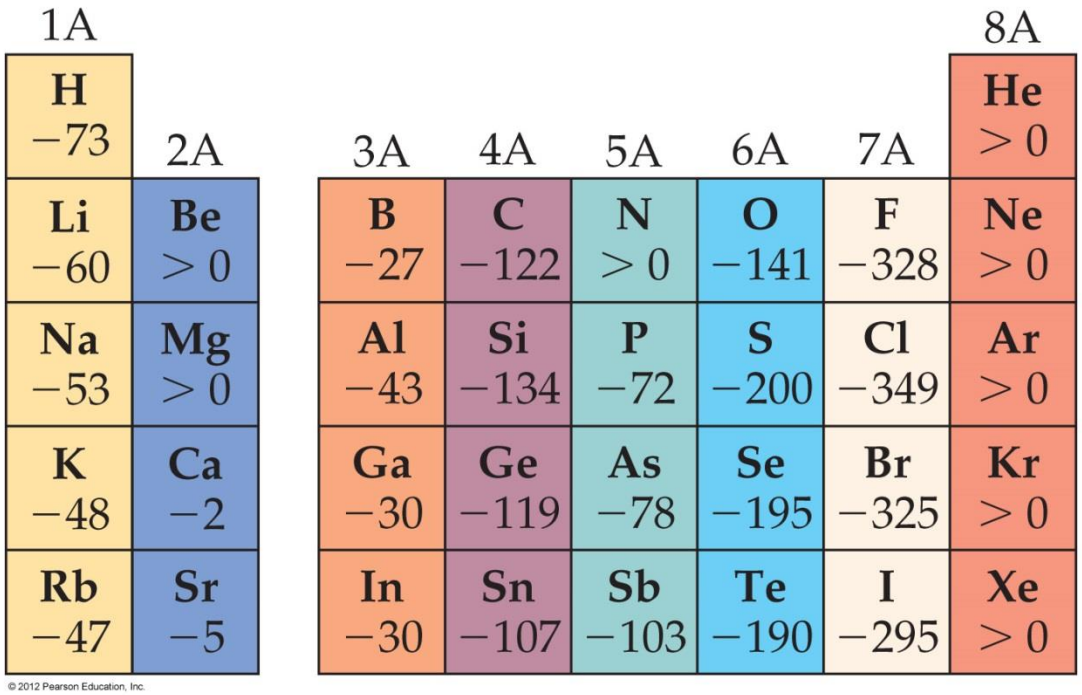
1A	2A	3A	4A	5A	6A	7A	8A
H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0

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There are again, however, two discontinuities in this trend.

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Trends in Electron Affinity



1A	2A	3A	4A	5A	6A	7A	8A
H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0

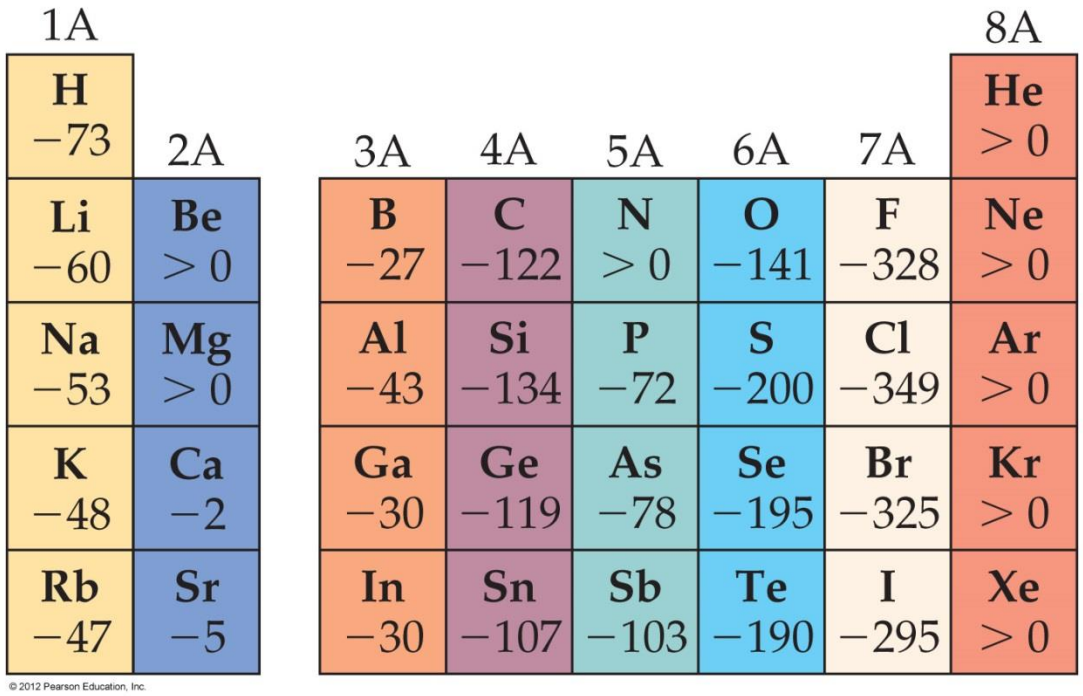
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The first occurs between Groups IA and IIA.

- The added electron must go in a *p* orbital, not an *s* orbital.
- The electron is farther from the nucleus and feels repulsion from the *s* electrons.

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Trends in Electron Affinity



1A	2A	3A	4A	5A	6A	7A	8A
H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0

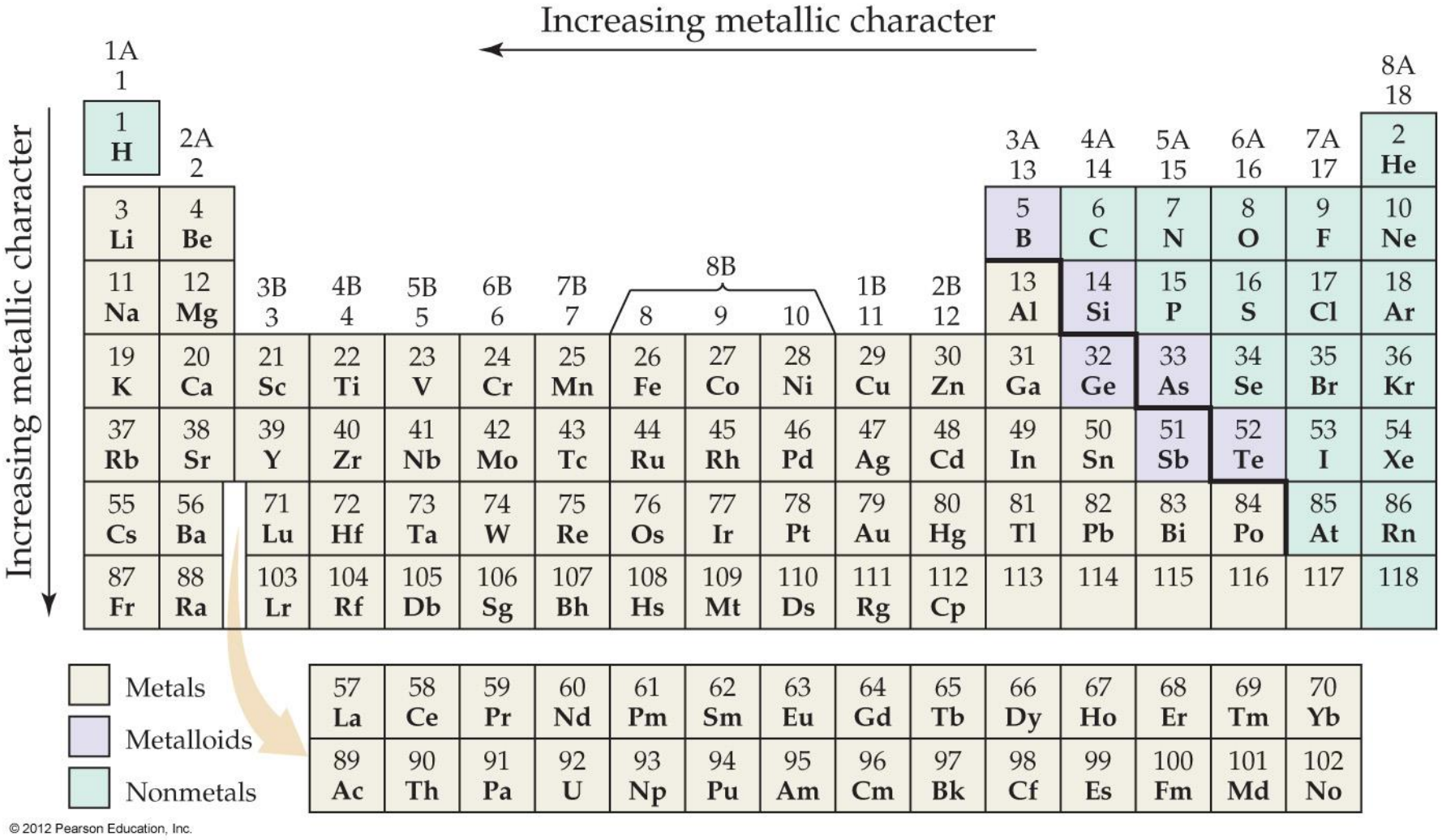
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The second discontinuity occurs between Groups IVA and VA.

- Group VA has no empty orbitals.
- The extra electron must go into an already occupied orbital, creating repulsion.

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Properties of Metal, Nonmetals,



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Periodic Properties of the Elements

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Metals versus Nonmetals

TABLE 7.3 • Characteristic Properties of Metals and Nonmetals

Metals	Nonmetals
Have a shiny luster; various colors, although most are silvery	Do not have a luster; various colors
Solids are malleable and ductile	Solids are usually brittle; some are hard, some are soft
Good conductors of heat and electricity	Poor conductors of heat and electricity
Most metal oxides are ionic solids that are basic	Most nonmetal oxides are molecular substances that form acidic solutions
Tend to form cations in aqueous solution	Tend to form anions or oxyanions in aqueous solution

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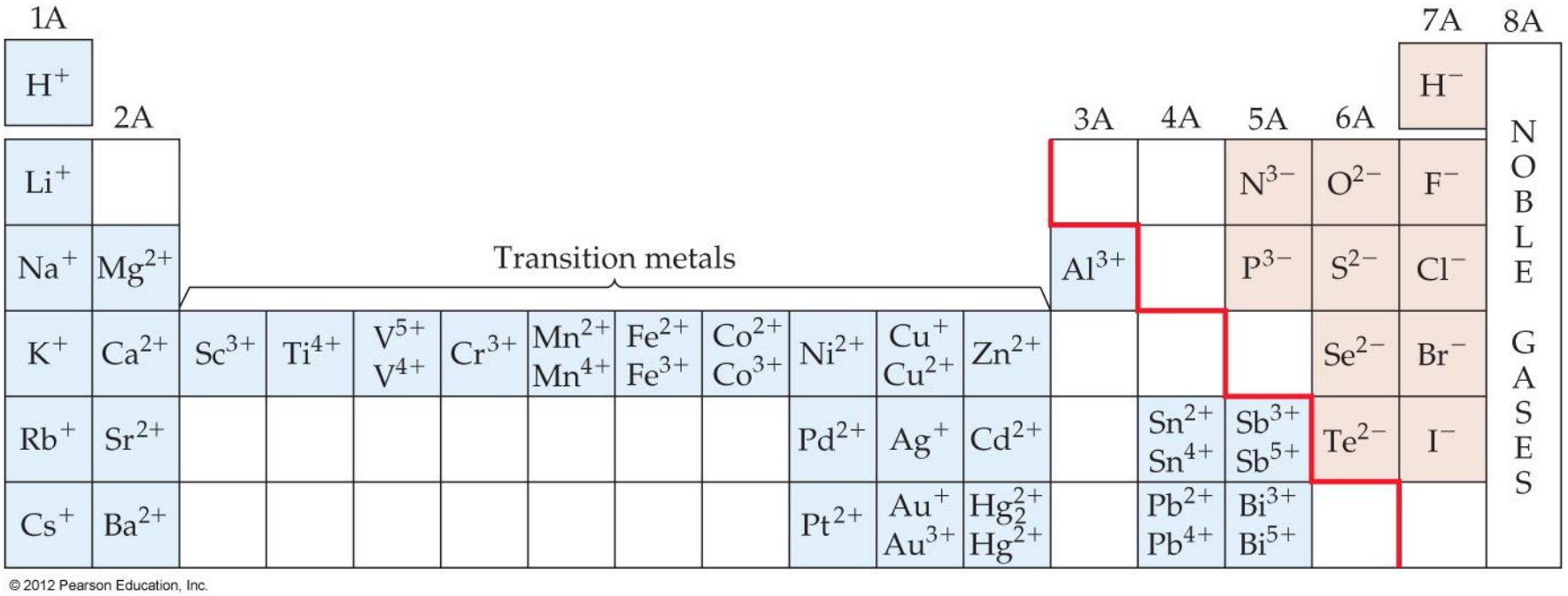
Differences between metals and nonmetals tend to revolve around these properties.

Periodic Properties of the Elements

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Metals versus Nonmetals

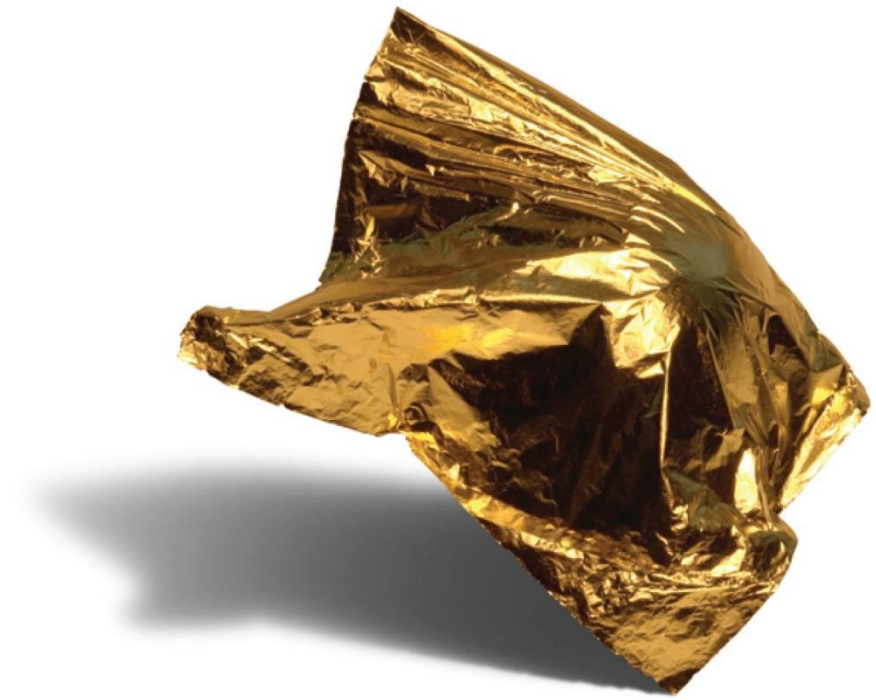
- Metals tend to form cations.
- Nonmetals tend to form anions.



Periodic Properties of the Elements

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Metals



Metals tend to be lustrous, malleable, ductile, and good conductors of heat and electricity.

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Metals

- Compounds formed between metals and nonmetals tend to be ionic.
- Metal oxides tend to be basic.



Nickle oxide (NiO), nitric acid (HNO_3), and water



NiO is insoluble in water but reacts with HNO_3 to give a green solution of the salt $Ni(NO_3)_2$

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Periodic Properties of the Elements

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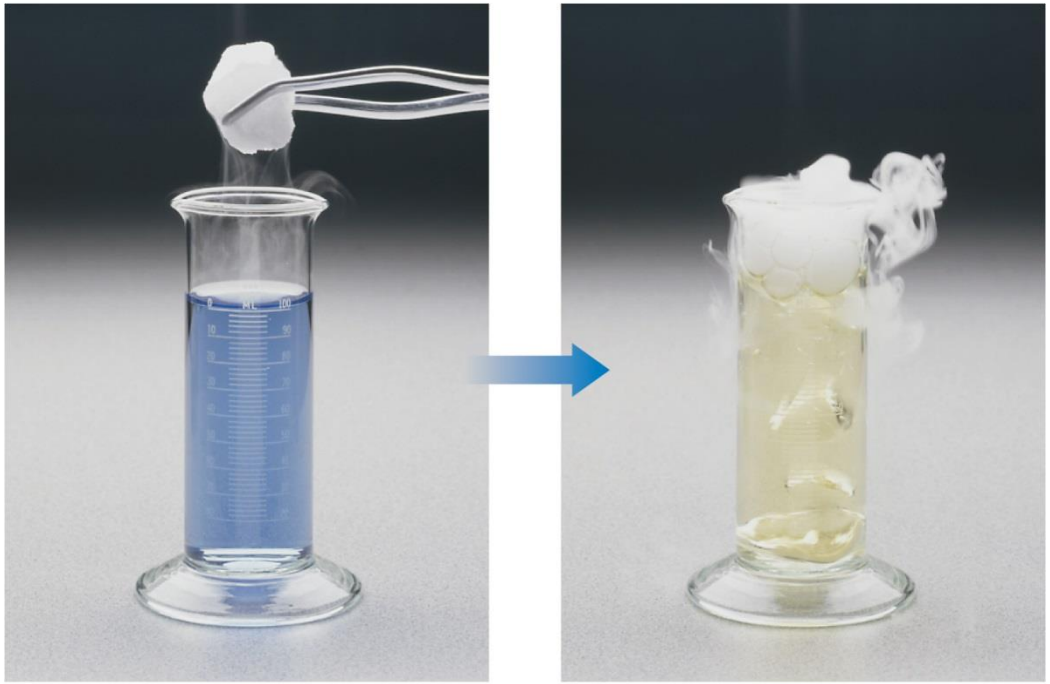
Nonmetals



- Nonmetals are dull, brittle substances that are poor conductors of heat and electricity.
- They tend to gain electrons in reactions with metals to acquire a noble-gas configuration.

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Nonmetals



- Substances containing only nonmetals are molecular compounds.
- Most nonmetal oxides are acidic.

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Metalloids



- Metalloids have some characteristics of metals and some of nonmetals.
- For instance, silicon looks shiny, but is brittle and a fairly poor conductor.
- boron, silicon, germanium, arsenic, antimony, tellurium, and polonium

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Group Trends

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Alkali Metals

- **Alkali metals** are soft, metallic solids.
- The name comes from the Arabic word for ashes.



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Alkali Metals

- They are found only in compounds in nature, not in their elemental forms.
- They have low densities and melting points.
- They also have low ionization energies.

TABLE 7.4 • Some Properties of the Alkali Metals

Element	Electron Configuration	Melting Point (°C)	Density (g/cm^3)	Atomic Radius (Å)	I_1 (kJ/mol)
Lithium	$[He]2s^1$	181	0.53	1.34	520
Sodium	$[Ne]3s^1$	98	0.97	1.54	496
Potassium	$[Ar]4s^1$	63	0.86	1.96	419
Rubidium	$[Kr]5s^1$	39	1.53	2.11	403
Cesium	$[Xe]6s^1$	28	1.88	2.25	376

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Alkali Metals



Li



Na



K

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Their reactions with water are famously exothermic.

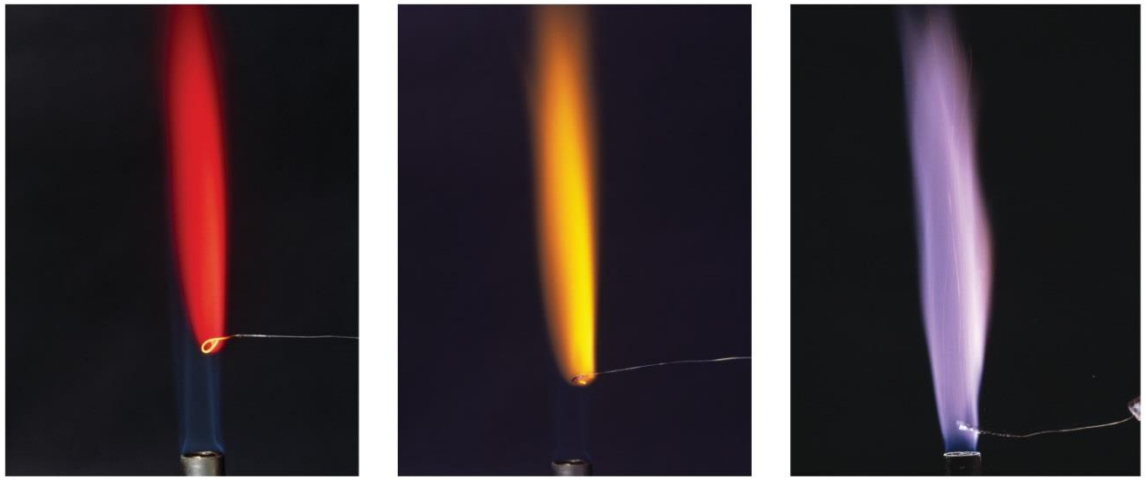
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Alkali Metals

- Alkali metals (except Li) react with oxygen to form peroxides.
- K, Rb, and Cs also form superoxides:

$$K + O_2 \longrightarrow KO_2$$

- They produce bright colors when placed in a flame.



Li Na K

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Alkaline Earth Metals

TABLE 7.5 • Some Properties of the Alkaline Earth Metals

Element	Electron Configuration	Melting Point (°C)	Density (g/cm^3)	Atomic Radius (Å)	I_1 (kJ/mol)
Beryllium	$[He]2s^2$	1287	1.85	0.90	899
Magnesium	$[Ne]3s^2$	650	1.74	1.30	738
Calcium	$[Ar]4s^2$	842	1.55	1.74	590
Strontium	$[Kr]5s^2$	777	2.63	1.92	549
Barium	$[Xe]6s^2$	727	3.51	1.98	503

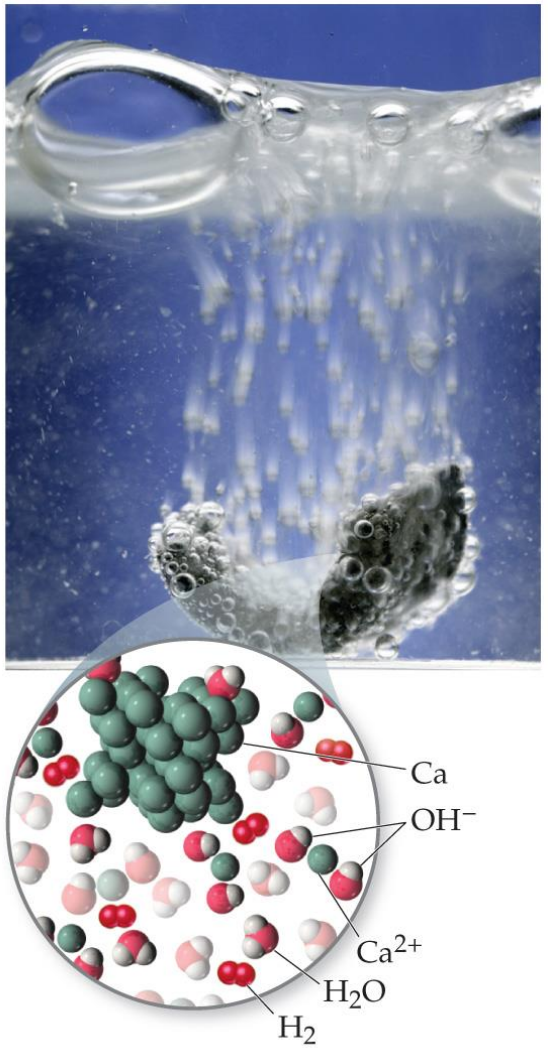
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- Alkaline earth metals have higher densities and melting points than alkali metals.
- Their ionization energies are low, but not as low as those of alkali metals.

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Alkaline Earth Metals

- Beryllium does not react with water, and magnesium reacts only with steam, but the other alkaline earth metals react readily with water.
- Reactivity tends to increase as you go down the group.



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Group 6A

TABLE 7.6 • Some Properties of the Group 6A Elements

Element	Electron Configuration	Melting Point (°C)	Density	Atomic Radius (Å)	I_1 (kJ/mol)
Oxygen	$[He]2s^2 2p^4$	−218	1.43 g/L	0.73	1314
Sulfur	$[Ne]3s^2 3p^4$	115	1.96 g/cm^3	1.02	1000
Selenium	$[Ar]3d^{10} 4s^2 4p^4$	221	4.82 g/cm^3	1.16	941
Tellurium	$[Kr]4d^{10} 5s^2 5p^4$	450	6.24 g/cm^3	1.35	869
Polonium	$[Xe]4f^{14} 5d^{10} 6s^2 6p^4$	254	9.20 g/cm^3	—	812

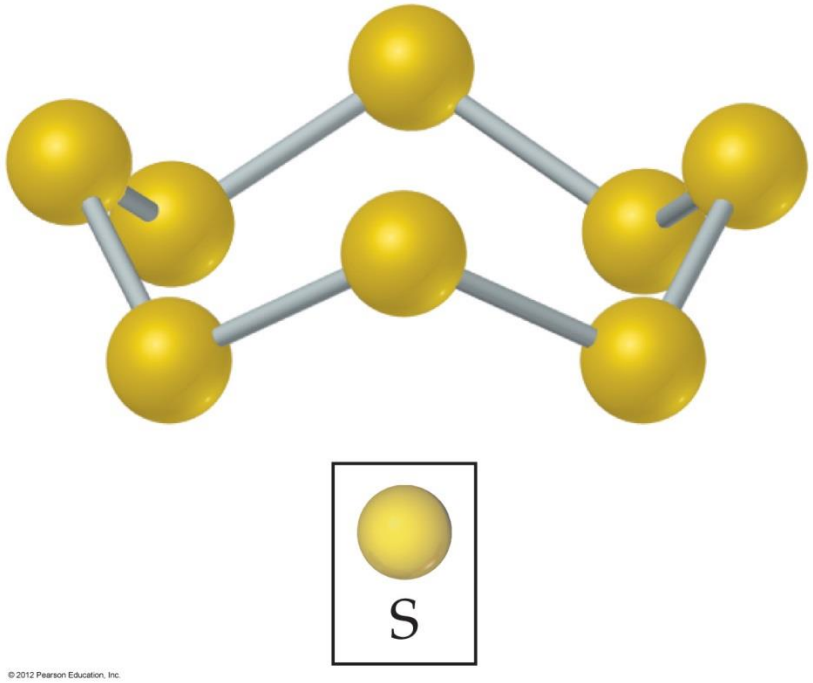
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- Oxygen, sulfur, and selenium are nonmetals.
- Tellurium is a metalloid.
- The radioactive polonium is a metal.

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Sulfur

- Sulfur is a weaker oxidizer than oxygen.
- The most stable allotrope is S_8, a ringed molecule.



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Group VIIA: Halogens

TABLE 7.7 • Some Properties of the Halogens

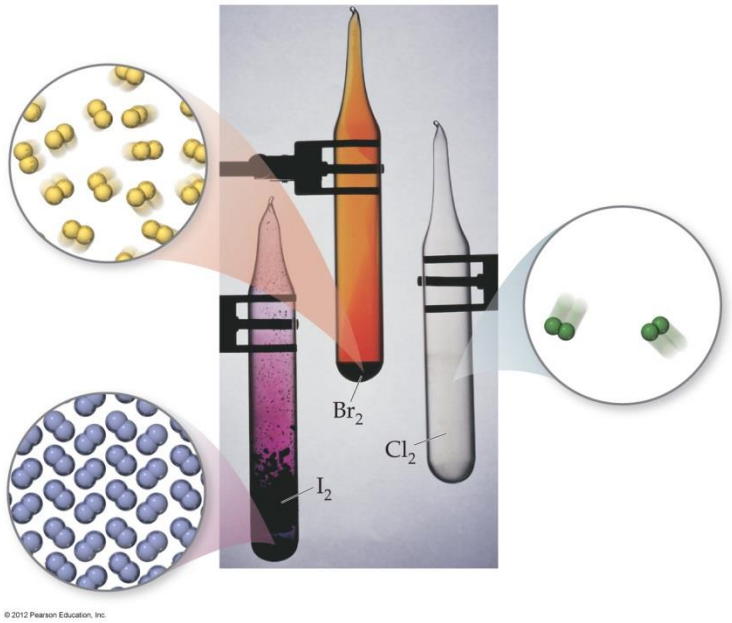
Element	Electron Configuration	Melting Point (°C)	Density	Atomic Radius (Å)	I_1 (kJ/mol)
Fluorine	$[He]2s^2 2p^5$	−220	1.69 g/L	0.71	1681
Chlorine	$[Ne]3s^2 3p^5$	−102	3.12 g/L	0.99	1251
Bromine	$[Ar]3d^{10} 4s^2 4p^5$	−7.3	3.12 g/cm^3	1.14	1140
Iodine	$[Kr]4d^{10} 5s^2 5p^5$	114	4.94 g/cm^3	1.33	1008

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- The **halogens** are prototypical nonmetals.
- The name comes from the Greek words *halos* and *gennao*: "salt formers."

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Group VIIA: Halogens



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- They have large, negative electron affinities.
  - Therefore, they tend to oxidize other elements easily.
- They react directly with metals to form metal halides.
- Chlorine is added to water supplies to serve as a disinfectant.

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Group VIIIA: Noble Gases

TABLE 7.8 • Some Properties of the Noble Gases

Element	Electron Configuration	Boiling Point (K)	Density (g/L)	Atomic Radius* (Å)	I_1 (kJ/mol)
Helium	$1s^2$	4.2	0.18	0.32	2372
Neon	$[He]2s^2 2p^6$	27.1	0.90	0.69	2081
Argon	$[Ne]3s^2 3p^6$	87.3	1.78	0.97	1521
Krypton	$[Ar]3d^{10} 4s^2 4p^6$	120	3.75	1.10	1351
Xenon	$[Kr]4d^{10} 5s^2 5p^6$	165	5.90	1.30	1170
Radon	$[Xe]4f^{14} 5d^{10} 6s^2 6p^6$	211	9.73	1.45	1037

*Only the heaviest of the noble-gas elements form chemical compounds. Thus, the atomic radii for the lighter noble-gas elements are estimated values.

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- The **noble gases** have astronomical ionization energies.
- Their electron affinities are positive.
  - Therefore, they are relatively unreactive.
- They are found as monatomic gases.

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